



MIDTERM STUDY GUIDE, FALL 2025

Scientific Habits of Mind Learning Objectives

STATISTICS (Smart Sparrow lessons 1&3, Tutorial)

1. Demonstrate an understanding of mean, standard deviation, and standard error.
2. Estimate mean and standard deviation from a distribution (histogram).
3. Calculate/estimate ~95% confidence intervals ($\text{mean} \pm 2\text{SE}$ intervals).
4. Determine statistical significance from confidence intervals and p-values (for bar graphs and correlations).
5. Distinguish between random and systematic errors and describe their respective implications for precision and accuracy.

CALCULATIONS (Smart Sparrow lessons 2&5, Tutorials)

6. Use units and unit conversions in calculations, utilize dimensional analysis.
7. Manipulate algebraic expressions and powers of ten.
8. Identify quantities of interest and applicable physics equations, use familiar and unfamiliar physics equations to calculate quantities of interest. Check answer plausibility (does your answer make sense?), put your answer into context by taking ratios/determining order of magnitude difference.
9. Identify when to use back-of-the-envelope calculations, make estimates using back-of-the-envelope calculations, identify and state assumptions, check answer plausibility.

SCIENTIFIC CLAIMS AND PROCESS (Smart Sparrow lesson 4, Tutorial)

10. Describe the role of falsifiability in science.
11. Distinguish between scientific hypotheses and scientific theories.
12. Consider experimental study design, determine the purpose of controls in an experiment, assess what makes a good control group or control task.
13. Distinguish between observations and experiments, identify strengths and weaknesses of experimental and observational studies.
14. Distinguish between correlation and causation. Identify potential explanations for an observed correlation, and evaluate whether a causal relationship is well supported.
15. Evaluate hypotheses in the context of presented evidence, propose experiments to test hypotheses or extend previous studies.
16. Read and interpret graphical information in histograms, bar graphs, scatter plots using axes labels, numerical information and units, error bars, and figure captions.
17. Consider the role of models and thought experiments in understanding physical processes, utilize models and thought experiments, identify their limitations.

SENSE OF SCALE (Smart Sparrow lessons 2&6, Tutorial)

18. Develop a sense of scale for the brain.
19. Demonstrate a sense of scale for speed, time, distance, and masses in the Universe. Use benchmarks and analogies to put quantities into context.
20. Demonstrate a sense of scale for instances in which (a) time, space, and energy for objects in relative motion are affected; and (b) wave-like behavior and limitations introduced by the Heisenberg Uncertainty Principle become important.
21. Utilize logarithmic scales, read values off a log scale, add values to a log scale. Identify when a log scale is useful, and when it cannot be used.

Concept Learning Objectives

Mind and Brain Week 1: Your Brain, What and How?

1. Describe the basic structure of a neuron, and explain how it facilitates communication with other neurons. Contrast how information flows within a neuron and between neurons.
2. Develop a working vocabulary for the macroscopic structure of the human brain: white and gray matter, hemispheres, corpus callosum, cortex, lobes.
3. Explain what it means that brain function is localized, and how different stimuli can lead to different perceptions (e.g. touch or hearing). Summarize the relationship between brain function, our experiences, and reality.
4. Compare and contrast three different techniques used to study the brain. Specify the assumptions and limitations that underlie fMRI measurements.
5. Consider why converging evidence is important for studying the relationship between mind and brain.

Mind and Brain Week 2: How Do We Remember?

1. Define the role of the hippocampus in memory, and articulate how we know using examples from lecture and reading.
2. Explain the terms brain plasticity and long-term potentiation.
3. List the functions, other than its role in memory, that the hippocampus appears to have, and describe the supporting evidence. Discuss the relevance of these additional functions of the hippocampus to the hypothesis that the hippocampus “binds elements of experience in time and space.”

Mind and Brain Week 3: How Do We Make Decisions?

1. Distinguish between habitual and deliberative decision-making.
2. Define reward prediction error, and explain how dopamine facilitates learning of habits and habitual behavior. Identify how we know and what part of the brain appears to be involved.
3. Describe an experiment that investigates deliberative decision-making.
4. Discuss applications of decision-making research, specifically using examples from lecture and readings.

Physics and Reality Week 1: Special Relativity

1. Describe the difference in speed, mass, and size scales that we experience in daily life vs. what exists in the Universe as a whole.
2. Explain why the observation that the speed of light is constant puzzled physicists in the late 19th and early 20th century and how Einstein resolved this puzzle.
3. Elaborate on the implications of the constancy of the speed of light for simultaneity and describe the example of the peace-signing treaty.
4. Evaluate qualitatively and quantitatively what happens to time and space for observers moving relative to one another at a constant velocity. Identify the conditions under which special relativity effects on time and space must be accounted for.
5. Summarize the significance of the equation $E=mc^2$ and the effects that moving fast has on mass.

Physics and Reality Week 2: General Relativity

1. Discuss the main features and limitations of Newton's Gravitational Law.
2. Describe Einstein's Theory of General Relativity qualitatively, and explain how it addresses the limitations of Newton's Gravitational Law.
3. Explain the Equivalence Principle and its importance to the development of Einstein's theory.
4. Enumerate the evidence in support of the Theory of General Relativity.
5. Summarize the main features of a black hole. Develop a sense of scale for black holes by calculating the Schwarzschild radius and/or mass of a black hole. Describe the effect that massive objects have on time (gravitational time dilation) qualitatively and quantitatively.
6. Explain what it means that the Universe is expanding. Interpret actual and hypothetical Hubble data, and determine what such data can tell us about the evolution of the Universe.

Physics and Reality Week 3: Quantum Mechanics

1. Describe what happens when the following objects pass through a wall with two-slits: a) pellets; b) water waves; c) electrons. Provide the interpretation of these results discussed in lecture.
2. Explain the role that probability plays in Quantum Mechanics.
3. Discuss the Heisenberg Uncertainty Principle and the way in which it fundamentally limits the precision of concurrent measurements of position and momentum/speed of an object.
4. Summarize what happens to a particle when measurement is undertaken on its entangled partner and what this tells us about the usual notion of locality.